

The Yield Strength of Spacetime:

Deriving the Planck Luminosity as the Rupture Limit of the Einstein Tiling

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Abstract

We reinterpret the Planck Luminosity ($L_p = c^5/G$) not as a theoretical abstraction, but as the physical breakdown voltage of the Axiomatic Physical Homeostasis (APH) lattice. We posit that Luminosity is the rate at which the vacuum can dissipate Associator Stress ($\mathcal{A}$) via the stiffness parameter β . We demonstrate that L_p represents the critical flux threshold where the geometric stiffness drops to zero ($\beta \rightarrow 0$) instantaneously. Above this limit, the Einstein Tiling cannot propagate information; the domain wall shatters, resulting in a localized *Big Bang* event or total metric failure. Thus, the Planck Luminosity is the upper bound of the *Geometric Stiffness Reactor* of the universe.

1 The Geometric Derivation

In the APH framework, energy flux Φ is the propagation of tiling defects through the Octonionic filament. The vacuum has a finite elastic modulus, defined by the inverse gravitational constant G^{-1} .

The maximum power the lattice can transmit is determined by the speed of causality c and the maximum tension the *Einstein Monotile* can support before the matching rules fail.

$$L_{max} = \frac{\text{Energy}_{rupture}}{t_{Planck}} = \frac{\beta_{crit} \cdot c^5}{G} \quad (1)$$

Where $\beta_{crit} \approx 1$ is the normalized stiffness of the associative manifold.

2 The Physical Interpretation

If a system attempts to radiate energy at $L > L_p$, the *wires* melt. The local curvature R becomes infinite, creating a Sedenion Bubble. The Associative property $[x, y, z] = 0$ is violated macroscopically.

- **Sub-Critical** ($L < L_p$): Stars, Quasars, Gamma Ray Bursts. The geometry holds.
- **Critical** ($L = L_p$): The Big Bang. The moment the Sedenion Bulk first crystallized into the Octonionic Filament.

Therefore, the Big Bang was not an explosion *in* space, but the *solidification* of space at the maximum possible cooling rate.